

Group B — Menstrual Cycle Analysis

Cytokines, Immune Cells, and Birth Control

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This document explores the **Group B (menstrual cycle)** dataset. We'll:

1. Load and join the data
2. Make summary tables with p-values
3. Visualise cytokine and immune cell differences by birth control and time point
4. Run regression models

1 1. Load and Join Data

```
# --- Sample IDs ---
sample_ids <- read_csv(file.path(data_dir, "group_b_menstruation",
                                "00_sample_ids_period.csv"),
                      show_col_types = FALSE)

# --- Participant metadata ---
metadata <- read_csv(file.path(data_dir, "group_b_menstruation",
                                "01_participant_metadata_period.csv"),
                    show_col_types = FALSE)

# --- Luminex cytokines (long format) ---
luminex <- read_csv(file.path(data_dir, "group_b_menstruation",
                                "02_luminex_period.csv"),
                  show_col_types = FALSE)

# --- Flow cytometry ---
flow <- read_csv(file.path(data_dir, "group_b_menstruation",
                            "03_flow_cytometry_period.csv"),
                show_col_types = FALSE)
```

1.1 Join everything together

```
# Join sample IDs with metadata (keep only .x versions, rename)
samples <- sample_ids %>%
  left_join(metadata, by = "pid", relationship = "many-to-one") %>%
  rename(arm = arm.x) %>%
  select(-arm.y)

# Widen cytokines (one column per cytokine)
# Clean names: replace hyphens with underscores in ALL column names
luminex_wide <- luminex %>%
  filter(limits == "not_censored") %>%
  select(sample_id, cytokine, conc) %>%
  pivot_wider(names_from = cytokine, values_from = conc)

# Replace hyphens in column names with underscores
names(luminex_wide) <- str_replace_all(names(luminex_wide), "-", "_")

cyt_names <- setdiff(names(luminex_wide), "sample_id")

# Join cytokine values onto samples
data_cyt <- samples %>%
  left_join(luminex_wide, by = "sample_id")

# Join flow cytometry
data_full <- data_cyt %>%
  left_join(flow, by = "sample_id")
```

```

glimpse(data_full)
#> Rows: 108
#> Columns: 25
#> $ pid <chr> "pid_12", "pid_04", "pid_26", "pid_08", "pid_16", "~
#> $ time_point <chr> "week_post", "week_post", "onset", "end_bleeding", ~
#> $ arm <chr> "birth_control", "birth_control", "birth_control", ~
#> $ sample_id <chr> "SAMP001", "SAMP002", "SAMP003", "SAMP004", "SAMP00~
#> $ age <dbl> 30, 31, 29, 30, 31, 33, 31, 29, 30, 34, 30, 29, 30,~
#> $ pcos_status <chr> "no disease", "pcos", "no disease", "no disease", "~
#> $ period_product <chr> "tampon", "pad", "tampon", "menstrual_cup", "pad", ~
#> $ sex <chr> "female", "female", "female", "female", "female", "~
#> $ IL_1a <dbl> 9.424212, 53.934976, 191.520622, 125.987758, 11.336~
#> $ IL_1b <dbl> 0.8890026, NA, 1.3386883, 11.3879657, NA, 370.90702~
#> $ MIG <dbl> 89.693983, 61.706487, NA, NA, 153.010108, 4.581469,~
#> $ IFNg <dbl> NA, NA, 2.949544, 4.777207, NA, 1.770491, 1.534639,~
#> $ IP_10 <dbl> 57.4500708, 33.4978474, 2.8166343, 0.9296425, 3.234~
#> $ MIP_3a <dbl> 6.5506137, 7.1456528, NA, 0.7100304, 30.2926570, 2.~
#> $ IL_6 <dbl> NA, NA, 28.6634758, 1.8298285, 0.4589363, 7.8777020~
#> $ TNFa <dbl> NA, NA, NA, 1.038203, NA, NA, 12.074762, NA, 2.9364~
#> $ live_cd19_negative <dbl> 158332, 105670, 310305, 738273, 168895, 333240, 117~
#> $ cd45_negative <dbl> 80327, 39124, 254821, 692237, 163627, 240537, 24718~
#> $ cd45_positive <dbl> 71141, 62616, 49552, 44545, 4739, 90305, 90939, 666~
#> $ neutrophils <dbl> 58727, 57525, 14106, 25732, 3275, 65707, 84360, 460~
#> $ non_neutrophils <dbl> 12412, 5089, 35445, 18813, 1463, 24596, 6576, 20623~
#> $ cd3_negative <dbl> 5839, 3078, 15124, 8681, 469, 14864, 4171, 9506, 56~
#> $ cd3_positive <dbl> 5802, 1765, 17358, 8933, 716, 8853, 1786, 8713, 897~
#> $ cd4_t_cells <dbl> 2740, 765, 9294, 5307, 332, 4896, 940, 4545, 4586, ~
#> $ cd8_t_cells <dbl> 2257, 248, 4367, 2994, 154, 2951, 531, 2799, 2963, ~

```

2 Summary Tables with P-values

2.1 Cytokine levels by birth control arm

```

# Helper: run a t-test for each cytokine and return a tidy table
cyt_arm <- data_cyt %>%
  pivot_longer(cols = all_of(cyt_names),
               names_to = "cytokine", values_to = "conc") %>%
  drop_na(conc, arm)

cyt_arm %>%
  group_by(cytokine, arm) %>%
  summarise(n = n(),
            mean_conc = mean(conc, na.rm = TRUE),
            sd_conc = sd(conc, na.rm = TRUE),
            .groups = "drop") %>%
  pivot_wider(names_from = arm,
              values_from = c(n, mean_conc, sd_conc),
              names_glue = "{arm}_{.value}") %>%

```

```
# add p-values from t-test
left_join(
  cyt_arm %>%
    group_by(cytokine) %>%
    summarise(p_value = t.test(conc ~ arm)$p.value, .groups = "drop"),
  by = "cytokine"
) %>%
mutate(across(where(is.numeric), ~ round(., 3))) %>%
kable(caption = "Cytokine concentrations by birth control arm (t-test)")
```

Table 1: Cytokine concentrations by birth control arm (t-test)

cytokine	birth_con- trol_n	no_birth_con- trol_n	birth_con- trol_mean_conc	no_birth_con- trol_mean_conc	birth_con- trol_sd_conc	no_birth_con- trol_sd_conc	p_value
IFNg	36	34	4.261	5.661	8.985	6.508	0.456
IL_1a	60	47	119.493	269.046	164.935	435.754	0.030
IL_1b	40	35	51.747	35.404	122.947	50.381	0.445
IL_6	38	31	20.466	22.758	50.196	93.271	0.903
IP_10	57	45	73.145	43.969	256.667	105.135	0.438
MIG	52	37	252.626	209.596	681.947	409.148	0.712
MIP_3a	53	41	13.916	15.891	37.239	42.773	0.815
TNFa	23	28	3.324	14.801	3.940	34.517	0.092

2.2 Cytokine levels by time point

```
cyt_time <- data_cyt %>%
  pivot_longer(cols = all_of(cyt_names),
    names_to = "cytokine", values_to = "conc") %>%
  drop_na(conc, time_point)

cyt_time %>%
  group_by(cytokine, time_point) %>%
  summarise(mean_conc = mean(conc, na.rm = TRUE),
    sd_conc = sd(conc, na.rm = TRUE),
    n = n(),
    .groups = "drop") %>%
  mutate(across(where(is.numeric), ~ round(., 3))) %>%
  kable(caption = "Cytokine concentrations by menstrual cycle time point")
```

Table 2: Cytokine concentrations by menstrual cycle time point

cytokine	time_point	mean_conc	sd_conc	n
IFNg	end_bleeding	6.310	10.590	25
IFNg	onset	5.280	7.240	26
IFNg	week_post	3.124	3.164	8
IFNg	week_prior	2.346	1.884	11
IL_1a	end_bleeding	313.141	373.422	27
IL_1a	onset	337.858	452.207	26
IL_1a	week_post	51.673	53.680	27

cytokine	time_point	mean_conc	sd_conc	n
IL_1a	week_prior	43.721	42.322	27
IL_1b	end_bleeding	29.295	49.091	26
IL_1b	onset	82.692	144.984	27
IL_1b	week_post	17.140	25.024	11
IL_1b	week_prior	11.467	14.646	11
IL_6	end_bleeding	40.359	110.413	27
IL_6	onset	13.917	27.704	27
IL_6	week_post	0.544	0.421	10
IL_6	week_prior	2.467	2.217	5
IP_10	end_bleeding	11.950	33.620	24
IP_10	onset	15.848	27.463	24
IP_10	week_post	147.322	372.243	27
IP_10	week_prior	55.670	96.542	27
MIG	end_bleeding	21.301	18.066	19
MIG	onset	28.599	60.108	16
MIG	week_post	303.725	618.153	27
MIG	week_prior	438.100	808.573	27
MIP_3a	end_bleeding	1.913	1.210	19
MIP_3a	onset	2.316	2.832	21
MIP_3a	week_post	19.993	49.764	27
MIP_3a	week_prior	28.307	51.414	27
TNFa	end_bleeding	7.708	13.917	20
TNFa	onset	15.014	38.202	21
TNFa	week_post	1.529	0.949	4
TNFa	week_prior	2.550	3.368	6

```
# ANOVA p-values for time point effect on each cytokine
cyt_time %>%
  group_by(cytokine) %>%
  nest() %>%
  mutate(anova_p = map_dbl(data, ~ aov(conc ~ time_point, data = .x) %>%
    summary() %>% pluck(1, "Pr(>F)", 1))) %>%
  select(cytokine, anova_p) %>%
  mutate(across(where(is.numeric), ~ round(., 4))) %>%
  kable(caption = "ANOVA p-values: does cytokine concentration vary by time point?")
```

Table 3: ANOVA p-values: does cytokine concentration vary by time point?

cytokine	anova_p
IL_1a	0.0001
IL_1b	0.0645
MIG	0.0382
IP_10	0.0570
MIP_3a	0.0492
IFNg	0.4954
IL_6	0.3490
TNFa	0.6228

2.3 2.3 PCOS status

```
cyt_pcos <- data_cyt %>%
  pivot_longer(cols = all_of(cyt_names),
               names_to = "cytokine", values_to = "conc") %>%
  drop_na(conc, pcos_status)

cyt_pcos %>%
  group_by(cytokine, pcos_status) %>%
  summarise(mean_conc = mean(conc, na.rm = TRUE),
            sd_conc = sd(conc, na.rm = TRUE),
            n = n(),
            .groups = "drop") %>%
  pivot_wider(names_from = pcos_status,
              values_from = c(mean_conc, sd_conc, n),
              names_glue = "{pcos_status}_{.value}") %>%
  left_join(
    cyt_pcos %>%
      group_by(cytokine) %>%
      summarise(p_value = t.test(conc ~ pcos_status)$p.value, .groups = "drop"),
    by = "cytokine"
  ) %>%
  mutate(across(where(is.numeric), ~ round(., 3))) %>%
  kable(caption = "Cytokine concentrations by PCOS status (t-test)")
```

Table 4: Cytokine concentrations by PCOS status (t-test)

cy-	no dis-		no dis-		no		
tokine	ease_mean_conc	pcos_mean_conc	ease_sd_conc	pcos_sd_conc	disease_n	pcos_n	p_value
IFNg	3.168	6.522	2.265	10.405	33	37	0.063
IL_1a	165.909	203.409	280.607	356.830	52	55	0.546
IL_1b	44.512	43.796	98.086	95.303	34	41	0.975
IL_6	16.868	25.739	41.990	92.044	33	36	0.604
IP_10	92.575	30.410	288.427	45.022	49	53	0.142
MIG	235.448	234.042	707.574	433.161	44	45	0.991
MIP_3a	20.853	9.197	55.308	12.587	45	49	0.174
TNFa	3.672	14.141	4.223	34.031	22	29	0.112

2.4 2.4 Flow cytometry cell counts by birth control arm

```
flow_joined <- flow %>%
  left_join(samples, by = "sample_id", relationship = "many-to-one")

# Cell types to test
cell_cols <- c("live_cd19_negative", "cd45_negative", "cd45_positive",
              "neutrophils", "non_neutrophils", "cd3_negative",
              "cd3_positive", "cd4_t_cells", "cd8_t_cells")

flow_joined %>%
```

```

pivot_longer(cols = all_of(cell_cols), names_to = "cell_type",
             values_to = "count") %>%
group_by(cell_type, arm) %>%
summarise(mean_count = mean(count, na.rm = TRUE),
          sd_count = sd(count, na.rm = TRUE),
          n = n(),
          .groups = "drop") %>%
pivot_wider(names_from = arm,
            values_from = c(mean_count, sd_count, n),
            names_glue = "{arm}_{.value}") %>%
left_join(
  flow_joined %>%
    pivot_longer(cols = all_of(cell_cols), names_to = "cell_type",
                 values_to = "count") %>%
    group_by(cell_type) %>%
    summarise(p_value = t.test(count ~ arm)$p.value, .groups = "drop"),
  by = "cell_type"
) %>%
mutate(across(where(is.numeric), ~ round(., 3))) %>%
kable(caption = "Immune cell counts by birth control arm (t-test)")

```

Table 5: Immune cell counts by birth control arm (t-test)

cell_type	birth_con- trol_mean_count	no_birth_con- trol_mean_count	birth_con- trol_sd_count	no_birth_con- trol_sd_count	birth_con- trol_n	no_birth_con- trol_n	p_value
cd3_neg- ative	9269.350	14906.021	8619.078	13656.662	60	48	0.015
cd3_posi- tive	7206.700	8849.062	4603.819	7504.520	60	48	0.188
cd45_neg- ative	213592.133	282587.792	197591.364	264786.552	60	48	0.137
cd45_pos- itive	90633.700	102016.708	68453.671	73926.757	60	48	0.413
cd4_t_cells	3873.383	4674.354	2815.268	3424.094	60	48	0.195
cd8_t_cells	2223.067	2694.792	1487.507	2632.831	60	48	0.272
live_cd19_neg- ative	309234.400	390799.583	197449.508	280663.683	60	48	0.092
neu- trophils	72492.317	76173.062	61947.188	60514.748	60	48	0.757
non_neu- trophils	18137.700	25838.708	12652.964	20504.558	60	48	0.026

3 3. Graphs

3.1 3.1 Cytokine boxplots — birth control arm

```

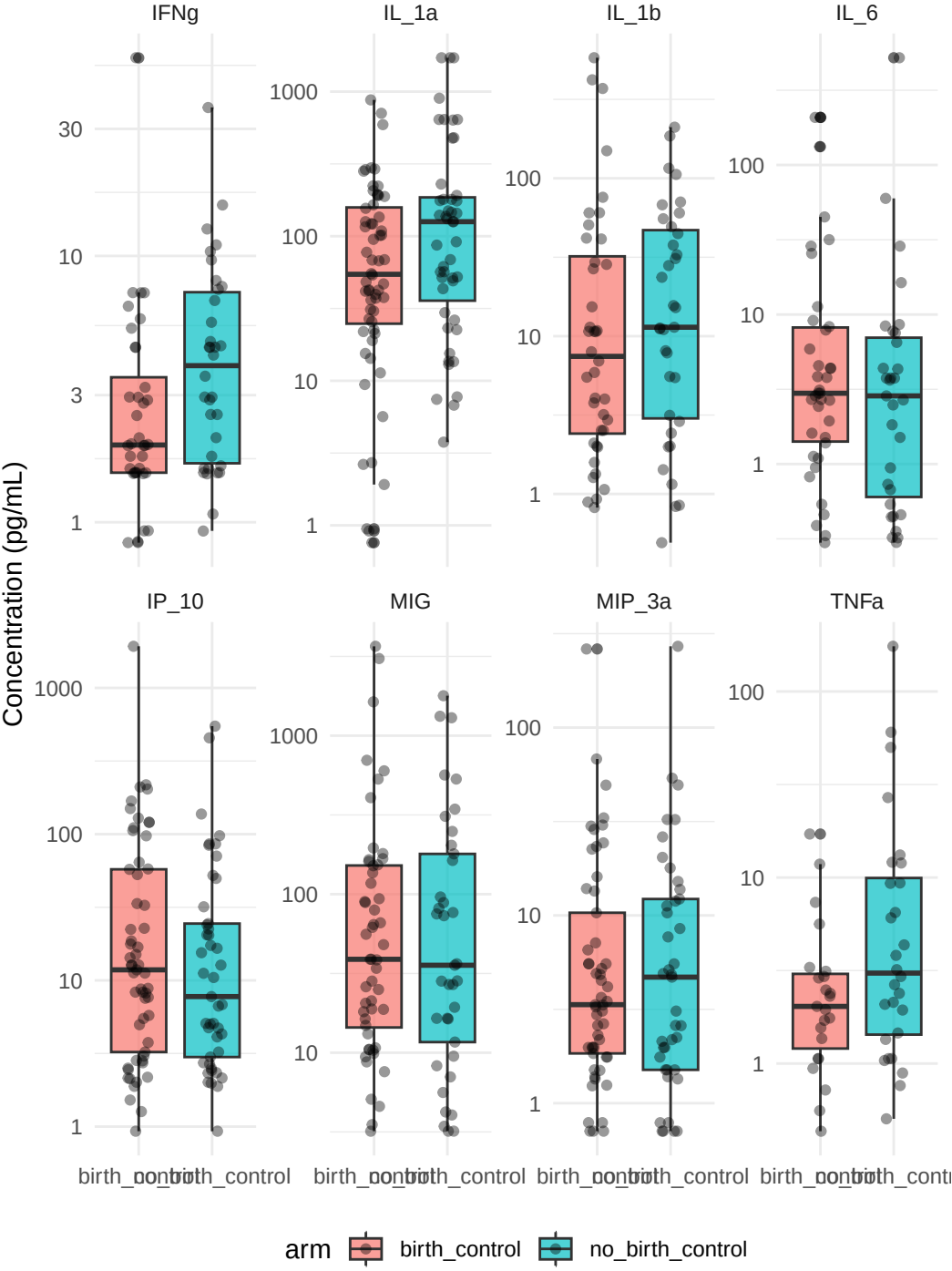
cyt_long <- data_cyt %>%
  pivot_longer(cols = all_of(cyt_names),
               names_to = "cytokine", values_to = "conc") %>%
  drop_na(conc, arm)

ggplot(cyt_long, aes(x = arm, y = conc, fill = arm)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 1.5) +
  facet_wrap(~ cytokine, scales = "free_y", ncol = 4) +
  scale_y_log10() +
  labs(title = "Cytokine concentrations by birth control arm",
       subtitle = "y-axis on log scale | jitter shows individual samples",
       x = NULL, y = "Concentration (pg/mL)") +
  theme_minimal() +
  theme(legend.position = "bottom")

```


Cytokine concentrations by birth control arm

y-axis on log scale | jitter shows individual samples



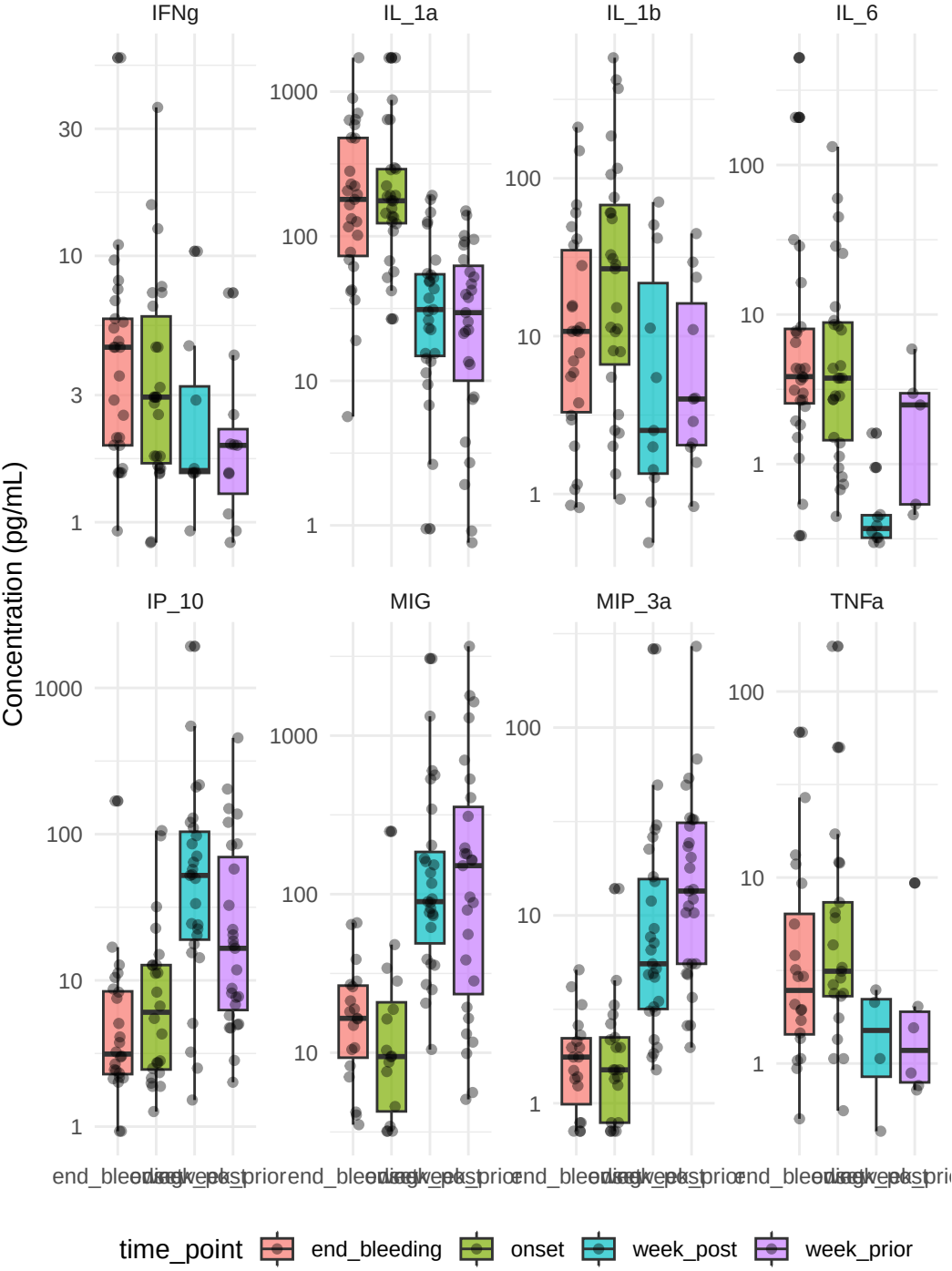
3.2 3.2 Cytokine boxplots — time point

```
cyt_long_time <- data_cyt %>%
  pivot_longer(cols = all_of(cyt_names),
               names_to = "cytokine", values_to = "conc") %>%
  drop_na(conc, time_point)

ggplot(cyt_long_time, aes(x = time_point, y = conc, fill = time_point)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 1.5) +
  facet_wrap(~ cytokine, scales = "free_y", ncol = 4) +
  scale_y_log10() +
  labs(title = "Cytokine concentrations across menstrual cycle time points",
       subtitle = "y-axis on log scale",
       x = NULL, y = "Concentration (pg/mL)") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

Cytokine concentrations across menstrual cycle time points

y-axis on log scale



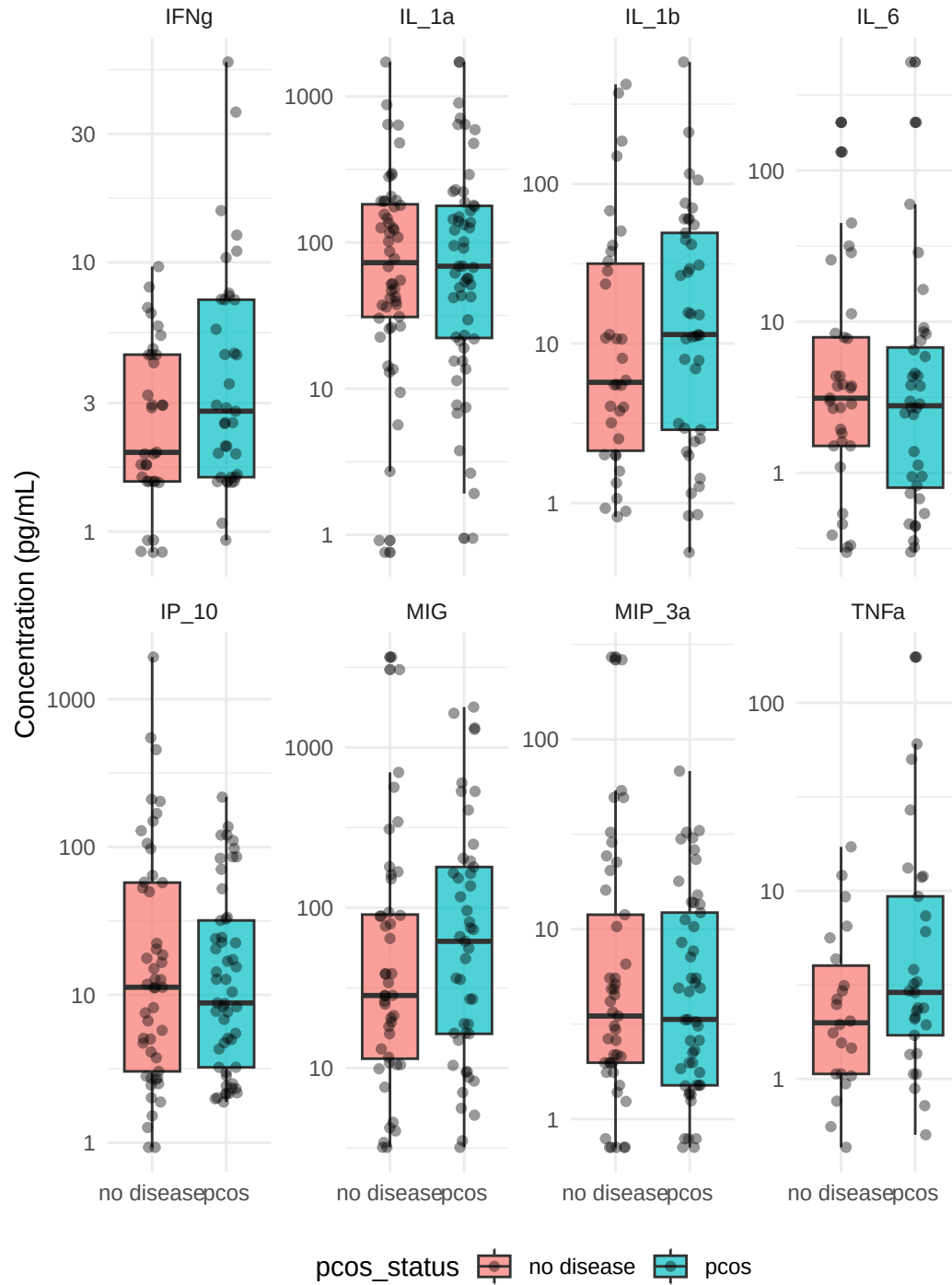
3.3 Cytokine boxplots — PCOS status

```
cyt_long_pcos <- data_cyt %>%
  pivot_longer(cols = all_of(cyt_names),
               names_to = "cytokine", values_to = "conc") %>%
  drop_na(conc, pcos_status)

ggplot(cyt_long_pcos, aes(x = pcos_status, y = conc, fill = pcos_status)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 1.5) +
  facet_wrap(~ cytokine, scales = "free_y", ncol = 4) +
  scale_y_log10() +
  labs(title = "Cytokine concentrations by PCOS status",
       subtitle = "y-axis on log scale",
       x = NULL, y = "Concentration (pg/mL)") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

Cytokine concentrations by PCOS status

y-axis on log scale



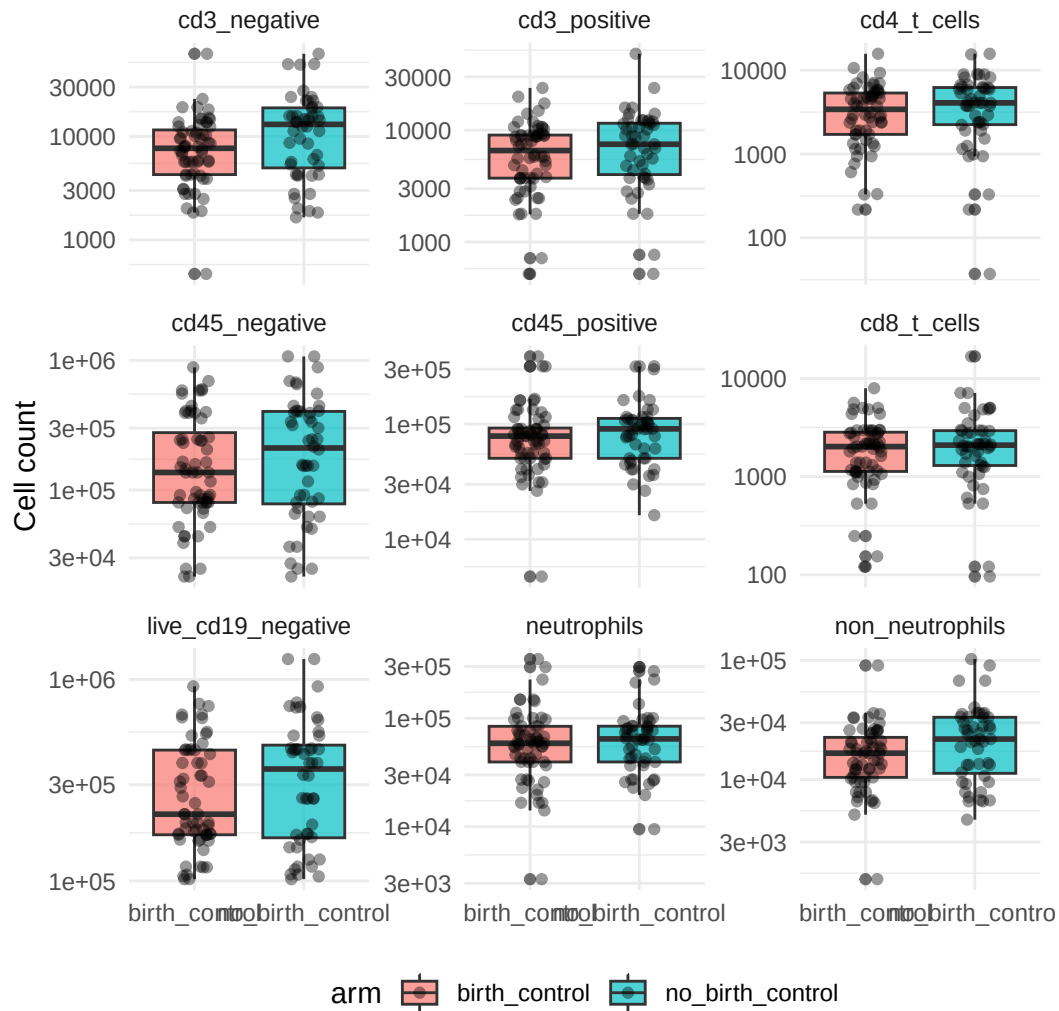
3.4 Immune cell counts by birth control arm

```
flow_long <- flow_joined %>%
  pivot_longer(cols = all_of(cell_cols), names_to = "cell_type",
               values_to = "count") %>%
  drop_na(count, arm)

ggplot(flow_long, aes(x = arm, y = count, fill = arm)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 1.5) +
  facet_wrap(~ cell_type, scales = "free_y", ncol = 3) +
  scale_y_log10() +
  labs(title = "Immune cell counts by birth control arm",
       subtitle = "y-axis on log scale",
       x = NULL, y = "Cell count") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

Immune cell counts by birth control arm

y-axis on log scale



3.5 Correlation heatmap — cytokines and immune cells

```
library(corrplot)

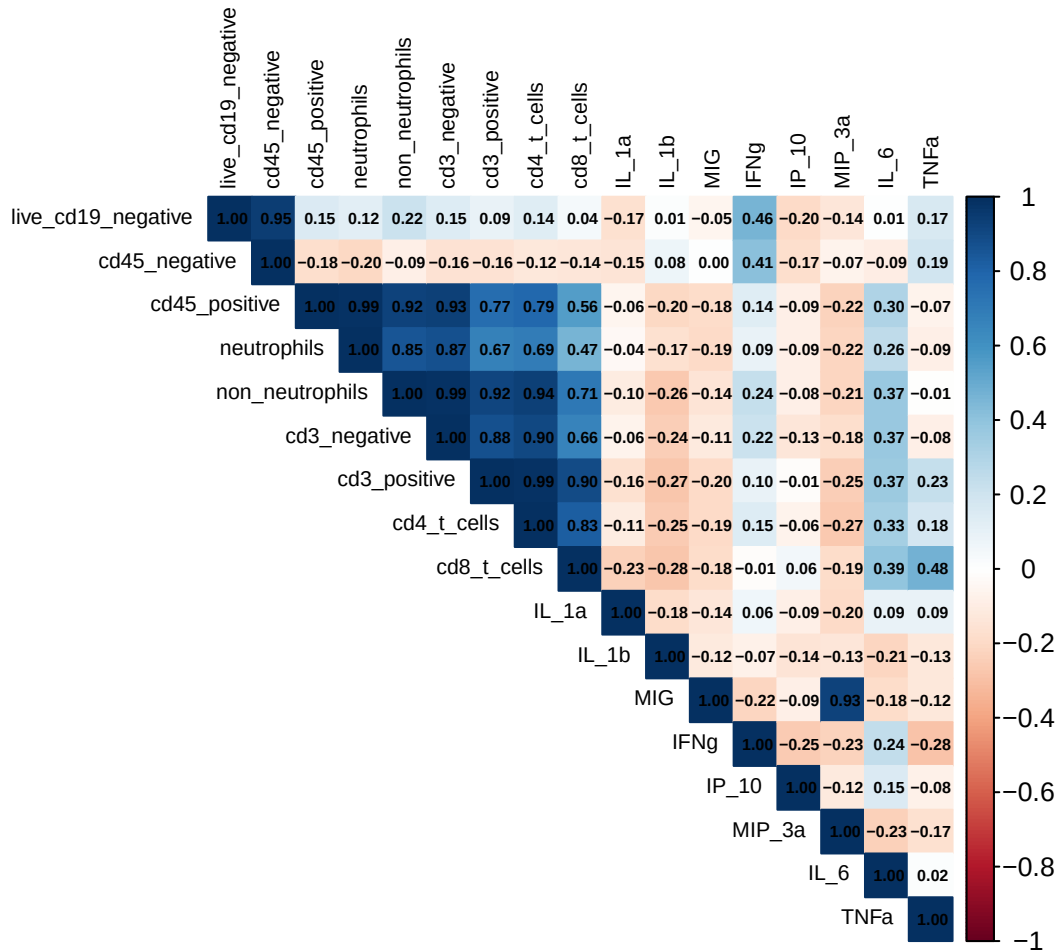
# Build a wide dataset with cytokines + key immune cells
corr_data <- data_full %>%
  select(all_of(cell_cols), all_of(cyt_names)) %>%
  drop_na()

cm <- cor(corr_data, use = "pairwise.complete.obs")

corrplot(cm, method = "color", type = "upper",
  tl.col = "black", tl.cex = 0.7,
  addCoef.col = "black", number.cex = 0.5,
```

```
title = "Correlation: Cytokines × Immune Cells",
mar = c(0, 0, 2, 0))
```

Correlation: Cytokines × Immune Cells

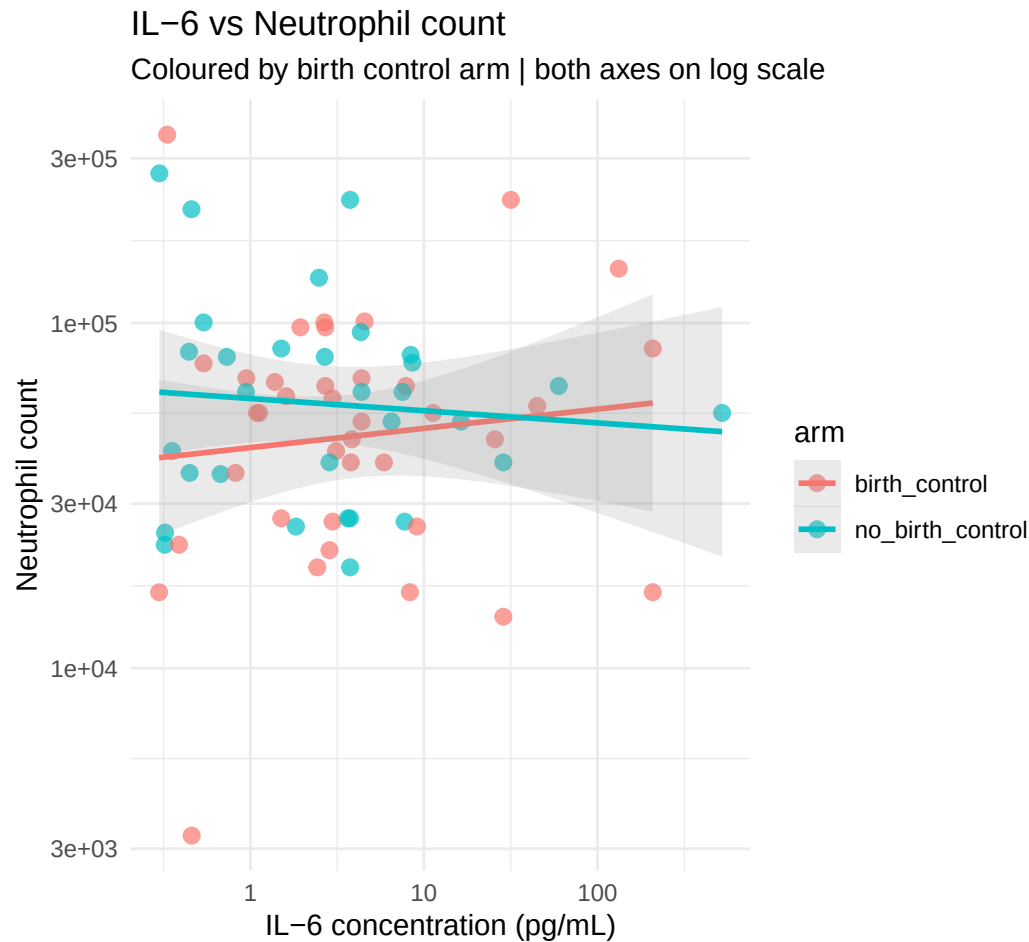


3.6 Scatter plot — strongest cytokine–immune cell relationship

```
data_full %>%
  drop_na(IL_6, neutrophils) %>%
  ggplot(aes(x = IL_6, y = neutrophils, color = arm)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.2) +
  scale_x_log10() +
  scale_y_log10() +
  labs(title = "IL-6 vs Neutrophil count",
       subtitle = "Coloured by birth control arm | both axes on log scale",
```



```
x = "IL-6 concentration (pg/mL)",
y = "Neutrophil count") +
theme_minimal()
```



4 4. Regression Models

4.1 4.1 Linear regression: cytokine ~ arm + age + pcos_status

We model each cytokine as a function of birth control arm, age, and PCOS status.

```
cyt_reg <- data_cyt %>%
  pivot_longer(cols = all_of(cyt_names),
               names_to = "cytokine", values_to = "conc") %>%
  drop_na(conc, arm, age, pcos_status) %>%
  mutate(conc_log = log10(conc + 1))

models <- cyt_reg %>%
  group_by(cytokine) %>%
```

```

nest() %>%
mutate(
  model = map(data, ~ lm(conc_log ~ arm + age + pcos_status, data = .x)),
  tidied = map(model, tidy)
) %>%
unnest(tidied) %>%
select(cytokine, term, estimate, std.error, statistic, p.value) %>%
mutate(across(where(is.numeric), ~ round(., 4)))

kable(models, caption = "Linear regression: log10(cytokine) ~ arm + age + pcos_status")

```

Table 6: Linear regression: $\log_{10}(\text{cytokine}) \sim \text{arm} + \text{age} + \text{pcos_status}$

cytokine	term	estimate	std.error	statistic	p.value
IL_1a	(Intercept)	2.3136	0.7507	3.0821	0.0026
IL_1a	armno_birth_control	0.2917	0.1317	2.2142	0.0290
IL_1a	age	-0.0184	0.0246	-0.7503	0.4548
IL_1a	pcos_statuspcos	-0.0454	0.1330	-0.3412	0.7337
IL_1b	(Intercept)	0.3473	0.8635	0.4022	0.6887
IL_1b	armno_birth_control	0.0405	0.1644	0.2463	0.8062
IL_1b	age	0.0230	0.0285	0.8079	0.4219
IL_1b	pcos_statuspcos	0.0709	0.1668	0.4249	0.6722
MIG	(Intercept)	2.1113	0.9167	2.3032	0.0237
MIG	armno_birth_control	-0.0559	0.1639	-0.3411	0.7339
MIG	age	-0.0152	0.0298	-0.5106	0.6110
MIG	pcos_statuspcos	0.2019	0.1632	1.2370	0.2195
IP_10	(Intercept)	1.2252	0.7321	1.6737	0.0974
IP_10	armno_birth_control	-0.0861	0.1361	-0.6326	0.5284
IP_10	age	0.0012	0.0240	0.0507	0.9597
IP_10	pcos_statuspcos	-0.0778	0.1370	-0.5682	0.5712
MIP_3a	(Intercept)	1.4881	0.5731	2.5965	0.0110
MIP_3a	armno_birth_control	0.0618	0.1079	0.5727	0.5683
MIP_3a	age	-0.0223	0.0188	-1.1850	0.2392
MIP_3a	pcos_statuspcos	-0.0196	0.1097	-0.1789	0.8584
IFNg	(Intercept)	0.7132	0.3936	1.8117	0.0746
IFNg	armno_birth_control	0.1153	0.0755	1.5261	0.1318
IFNg	age	-0.0060	0.0129	-0.4651	0.6434
IFNg	pcos_statuspcos	0.1001	0.0754	1.3288	0.1885
IL_6	(Intercept)	0.6700	0.7790	0.8601	0.3929
IL_6	armno_birth_control	-0.1104	0.1505	-0.7336	0.4658
IL_6	age	0.0038	0.0254	0.1507	0.8807
IL_6	pcos_statuspcos	-0.0312	0.1516	-0.2057	0.8377
TNFa	(Intercept)	-1.6925	0.6351	-2.6651	0.0105
TNFa	armno_birth_control	0.1393	0.1120	1.2441	0.2196
TNFa	age	0.0712	0.0209	3.4140	0.0013
TNFa	pcos_statuspcos	0.0889	0.1123	0.7911	0.4329

Interpretation: The `arm` coefficient tells us the difference in log-cytokine concentration between the `birth_control` and `no_birth_control` groups, holding age and PCOS status constant. A significant p-value (< 0.05) suggests the birth control arm has a meaningful association with that cytokine.

4.2 4.2 Linear regression: immune cells ~ arm + age + pcos_status

```
flow_reg <- flow_joined %>%
  pivot_longer(cols = all_of(cell_cols), names_to = "cell_type",
               values_to = "count") %>%
  drop_na(count, arm, age, pcos_status) %>%
  mutate(count_log = log10(count + 1))

models_flow <- flow_reg %>%
  group_by(cell_type) %>%
  nest() %>%
  mutate(
    model = map(data, ~ lm(count_log ~ arm + age + pcos_status, data = .x)),
    tidied = map(model, tidy)
  ) %>%
  unnest(tidied) %>%
  select(cell_type, term, estimate, std.error, statistic, p.value) %>%
  mutate(across(where(is.numeric), ~ round(., 4)))

kable(models_flow,
      caption = "Linear regression: log10(cell count) ~ arm + age + pcos_status")
```

Table 7: Linear regression: log10(cell count) ~ arm + age + pcos_status

cell_type	term	estimate	std.error	statistic	p.value
live_cd19_negative	(Intercept)	5.6975	0.3117	18.2795	0.0000
live_cd19_negative	armno_birth_control	0.1052	0.0575	1.8282	0.0704
live_cd19_negative	age	-0.0084	0.0102	-0.8211	0.4135
live_cd19_negative	pcos_statuspcos	-0.0705	0.0581	-1.2132	0.2278
cd45_negative	(Intercept)	5.4342	0.4786	11.3532	0.0000
cd45_negative	armno_birth_control	0.1402	0.0883	1.5872	0.1155
cd45_negative	age	-0.0073	0.0157	-0.4678	0.6409
cd45_negative	pcos_statuspcos	-0.1464	0.0893	-1.6400	0.1040
cd45_positive	(Intercept)	4.8483	0.3175	15.2689	0.0000
cd45_positive	armno_birth_control	0.0368	0.0586	0.6279	0.5314
cd45_positive	age	-0.0001	0.0104	-0.0088	0.9930
cd45_positive	pcos_statuspcos	0.0536	0.0592	0.9050	0.3675
neutrophils	(Intercept)	4.7182	0.3537	13.3381	0.0000
neutrophils	armno_birth_control	0.0224	0.0653	0.3432	0.7321
neutrophils	age	-0.0001	0.0116	-0.0067	0.9947
neutrophils	pcos_statuspcos	0.0635	0.0660	0.9626	0.3380
non_neutrophils	(Intercept)	4.0291	0.3247	12.4091	0.0000
non_neutrophils	armno_birth_control	0.1061	0.0599	1.7706	0.0796
non_neutrophils	age	0.0045	0.0106	0.4209	0.6747
non_neutrophils	pcos_statuspcos	0.0320	0.0605	0.5282	0.5985
cd3_negative	(Intercept)	3.5148	0.4049	8.6799	0.0000
cd3_negative	armno_birth_control	0.1292	0.0747	1.7286	0.0868
cd3_negative	age	0.0100	0.0133	0.7509	0.4544
cd3_negative	pcos_statuspcos	0.0647	0.0755	0.8568	0.3935
cd3_positive	(Intercept)	3.8169	0.3779	10.0989	0.0000
cd3_positive	armno_birth_control	0.0702	0.0697	1.0058	0.3169

cell_type	term	estimate	std.error	statistic	p.value
cd3_positive	age	-0.0016	0.0124	-0.1264	0.8997
cd3_positive	pcos_statuspcos	-0.0230	0.0705	-0.3267	0.7445
cd4_t_cells	(Intercept)	3.4955	0.4729	7.3921	0.0000
cd4_t_cells	armno_birth_control	0.0528	0.0873	0.6046	0.5468
cd4_t_cells	age	-0.0012	0.0155	-0.0774	0.9385
cd4_t_cells	pcos_statuspcos	-0.0014	0.0882	-0.0156	0.9876
cd8_t_cells	(Intercept)	3.2853	0.4112	7.9886	0.0000
cd8_t_cells	armno_birth_control	0.0454	0.0759	0.5982	0.5510
cd8_t_cells	age	-0.0013	0.0135	-0.0947	0.9248
cd8_t_cells	pcos_statuspcos	-0.0145	0.0767	-0.1887	0.8507

5 5. Quick Takeaways

```
# Count significant arm effects in cytokine regression
sig_cyt <- models %>%
  filter(term == "armbirth_control", p.value < 0.05) %>%
  pull(cytokine)

sig_flow <- models_flow %>%
  filter(term == "armbirth_control", p.value < 0.05) %>%
  pull(cell_type)

cat("**Cytokines significantly associated with birth control (p < 0.05):** \n")
```

Cytokines significantly associated with birth control (p < 0.05):

```
if (length(sig_cyt) > 0) {
  cat(paste0("- ", sig_cyt, collapse = "\n"))
} else {
  cat("- None (no significant associations detected)")
}
```

- None (no significant associations detected)

```
cat("\n\n**Immune cell types significantly associated with birth control (p < 0.05):** \n")
```

Immune cell types significantly associated with birth control (p < 0.05):

```
if (length(sig_flow) > 0) {
  cat(paste0("- ", sig_flow, collapse = "\n"))
} else {
  cat("- None (no significant associations detected)")
}
```

- None (no significant associations detected)